Week 4 Bayesian approach for classification?

- we can sample from the joint of labels y_n and inputs x_n

$$x^{(i)}, y^{(i)} \sim p(x, y)$$

Task 1.1 Implement a function evaluating the posterior distribution $p(y=1|x)$ and the marginal data distribution

$$\boxed{p(y=1|x) = \frac{p(x|y=1)\,p(y=1)}{p(x)} = \frac{\mathcal{N}(x|\mu_1,\Sigma_1)\cdot\tilde{\pi}_1}{\tilde{\pi}_0\mathcal{N}(x|\mu_0,\Sigma_1)+\tilde{\pi}_1\mathcal{N}(x|\mu_1,\Sigma_1)}}$$

Understanding laplace approximation

- the setup
  - N = # of trials (coin flips)
  - y = # of successes (heads)
  - α_0, β_0 Beta prior parameters
  - generative model $y|\theta \sim Bin(N=12, \theta)$ $\theta \sim Beta(1,1)$

- the joint model

$$P(y,\theta) = p(y|\theta)\,p(\theta) = \underbrace{\binom{N}{y}}_{\text{constant in } \theta} \overbrace{\theta^{y}(1-\theta)^{N-y}}^{\text{likelihood}} \underbrace{\frac{1}{Beta(\alpha_0,\beta_0)}}_{\text{constant in } \theta} \overbrace{\theta^{(\alpha_0-1)}(1-\theta)^{(\beta_0-1)}}^{\text{Prior}}$$

$$= \left[\binom{N}{y} B(\alpha_0,\beta_0)^{-1}\right] \theta^{y+\alpha_0-1}(1-\theta)^{N-y+\beta_0-1}$$

□ taking a logarithm

$$\log p(y,\theta) = \underbrace{\log\binom{N}{y} - \log B(\alpha_0,\beta_0)}_{\text{constant in } \theta = k} + (y+\alpha_0-1)\cdot\log\theta + (N-y+\beta_0-1)(1-\theta)$$

□ Plugging in numbers

$$f(\theta) = \log p(y,\theta) = \underbrace{(\overset{y}{2}+\overset{\alpha_0}{1}-1)}_{2}\log\theta + \underbrace{(\overset{N}{12}-\overset{y}{2}+\overset{\beta_0}{1}-1)}_{10}\log(1-\theta)+k = 2\log\theta + 10\log(1-\theta)+k$$

□ Taking the 1-st derivative

$$\frac{d}{d\theta}\ln p(y,\theta) = (\alpha_0+y-1)\frac{1}{\theta} - (\beta_0+N-y-1)\frac{1}{1-\theta}$$

- setting $\frac{d}{d\theta}=0$ and plugging in

$$0 = (1+2-1)\frac{1}{\theta} - (1+12-2-1)\frac{1}{1-\theta}$$

$$0 = \frac{2}{\theta} - \frac{10}{(1-\theta)}$$

$$0 = 2(1-\theta) - 10\theta$$

$$0 = 2 - 2\theta - 10\theta$$

$$2 = 12\theta \Rightarrow \boxed{\theta = 0,16} \boxed{= \theta_{MAP}}$$

□ alternatively we can use the closed form for Beta Binomial distribution to find the

$$\boxed{\theta_{MAP} = \frac{\alpha_0 + y - 1}{\alpha_0 + \beta_0 + N - 2}}$$

□ Taking the 2nd derivative to measure the curvature (Hessian)

$$\frac{d^2}{d\theta^2} = \ln p(y,\theta) = -(\alpha_0+y-1)\frac{1}{\theta^2} - (\beta_0+N-y-1)\frac{1}{(1-\theta)^2}$$

- plugging in $= -(1+2-1)\frac{1}{\theta^2} - (1+12-2-1)\frac{1}{(1-\theta)^2}$

$$= -\frac{2}{\theta^2} - \frac{10}{(1-\theta)^2}$$

△ evaluating the second derivative at $\theta_{MAP} = 0,16$

$$= -\frac{2}{(0,16)^2} - \frac{10}{(1-0,16)^2} = -72 - 14.4 = \boxed{-86.4}$$

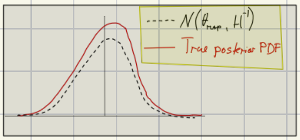
Laplace approximation: the idea is:

$$\log p(\theta|y) \approx \log p(\overset{\text{max. a. post. approx}}{\theta_{MAP}}|y) = -\frac{1}{2}(\theta-\theta_{MAP})^T \overset{\text{Hessian of negative log posterior evaluated at } \theta_{MAP}}{H} (\theta-\theta_{MAP})$$

$H = -\nabla^2 \log p(\theta|y)|_{\theta=\theta_{MAP}}$

- we are building a quadratic approx. of $\log p(\theta|y)$ at it's peak
- A large negative second derivative (-86.4) means our function is sharply peaked.
- laplace approx. uses this value to define the covariance of the Gaussian $Cov[\theta] \approx -H^{-1}$

  - in our case: $\theta \sim N(\theta_{MAP}, (-H)^{-1}) = N\left(\frac{1}{6}, \frac{1}{86.4}\right)$



- Converting curvature to a Gaussian variance

- for univariate Lap app.

$$q(\theta) = \mathcal{N}(\theta_{MAP}, A^{-1}), \text{ with } A = -\frac{d^2 f}{d\theta^2}\Big|_{\theta=\theta_{MAP}}$$

- here $A = 86.4 \Rightarrow \sigma^2 = \mathbb{V}[q] = A^{-1} = \frac{1}{86.4} \approx 0.01157$

$$\therefore\ q(\theta) = \mathcal{N}(0.16, 0.01157)$$

- Comparing w/ the **true** conjugate posterior

$$P(\theta|y) = \text{Beta}(\alpha_0+y, \beta_0+N-y) = \text{Beta}(3, 11)$$

- mean
  - exact Beta mode $\Rightarrow \theta_{mode} = \frac{\alpha-1}{\alpha+\beta-2} = \frac{(\alpha_0+y)-1}{(\alpha_0+y)+(\beta_0+N-y)-2} = \frac{1+2-1}{1+2+1+12-2-2} = 0{,}1667$
  - Laplace $= 0{,}1667$
- variance
  - exact Beta variance $\mathbb{V}[\theta] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{(\alpha_0+y)(\beta_0+N-y)}{((\alpha_0+y)(\beta_0+N-y))^2 \cdot ((\alpha_0+y)(\beta_0+N-y)+1)} = 0.01122$
  - Laplace $= 0{,}01157$

Logistic Regression Laplace Approximation

dataset $= \{(-2,0), (-1,0), (1,1), (2,1)\}$

- design matrix $X = \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, shape $(N \times D) = (4 \times 2)$ — # of observations, num of parameters
- targets: $y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ shape $(N \times 1) = 4 \times 1$
- we use it in the code $f_n = w^T x_n \Rightarrow f = Xw$
- Parameters vector w

  Because $D = 2$

$$w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

intercept (w_0), slope w.r.t. x (w_1)

- Prior
  - Precision $\alpha = 1$

$$w \sim \mathcal{N}(0, \alpha^{-1} I_2) = \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)$$

🧠 Before seeing the data we expect the intercept and the slope to be around 0.

- Gaussian pdf $p(w) = (2\pi)^{-\frac{D}{2}} \alpha^{\frac{D}{2}} \exp\left(-\frac{\alpha}{2} w^T w\right)$
- log prior $\log p(w) = -\frac{\alpha}{2} w^T w - \frac{D}{2}\log(2\pi) + \frac{D}{2}\log\alpha$

- Likelihood

$\forall n \quad y | w, x_n \sim \text{Bernoulli}(p_n), \quad p_n \equiv \sigma(f_n), \quad f_n = w^T x_n$

- where $\sigma(z) = \frac{1}{1+e^{-z}}$, Bernoulli pmf: $p(y_n | w) = \sigma(f_n)^{y_n} [1 - \sigma(f_n)]^{1-y_n}$ $y_n \in \{0,1\}$

- joint likelihood for the whole data

$$p(y|w) = \prod_{n=1}^{N} \sigma(f_n)^{y_n} \left[1-\sigma(f_n)\right]^{1-y_n}$$

- taking logarithm

$$\log p(y|w) = \sum_{n=1}^{N} \left[y_n \log \sigma(f_n) + (1-y_n) \log(1-\sigma(f_n)) \right]$$

- Combine log-joint $p(y,w) = p(y|w)\,p(w)$

$$\log p(y,w) = \underbrace{\sum_{n=1}^{N} \left[y_n \log \sigma(\overset{= w^T x_n}{f_n}) + (1-y_n) \log(1-\sigma(\overset{= w^T x_n}{f_n})) \right]}_{\text{likelihood}} \underbrace{- \frac{\alpha}{2} w^T w}_{\text{prior}} + C = -\frac{D}{2}\log(2\pi) + \frac{D}{2}\log\alpha$$

- derivative picking a single data pt. n and a single coordinate k of the weight vector, the n-th summand is

$$\ell_n = y_n \log \sigma_n + (1-y_n)\log(1-\sigma_n)$$

- we want $\dfrac{d\ell_n}{dw_k}$; sigmoid derivative $\dfrac{d\sigma(z)}{dz} = \sigma(z)[1-\sigma(z)]$, $f_n = \sum_{j=1}^{D} w_j x_{nj} \Rightarrow \dfrac{df_n}{dw_k} = x_{nk}$

chain rule for a composite $g(h(w))$ $\quad \dfrac{dg}{dw} = g'(h)\,h'(w)$

- derivative of the positive-class term

$$\frac{d}{dw_k} = \left[y_n \log \sigma_n\right] = y_n \frac{1}{\sigma_n} \frac{d\sigma_n}{dw_k} = y_n \frac{1}{\sigma_n} \sigma_n(1-\sigma_n)\,x_{nk} = y_n(1-\sigma_n)\,x_{nk}$$

- Gradient ∧ Hessian

  - $\nabla_w \log P(y, w) = X^T(y - \sigma) - \alpha w$
  - $\nabla^2_w \log P(y, w) = -X^T \operatorname{diag}(\sigma \odot (1-\sigma)) X - \alpha I$

    $(\sigma = \sigma(Xw))$

- finding map numerically

  $\hat{w}_{MAP} = \begin{bmatrix} 0.000 \\ 1.006 \end{bmatrix} \Rightarrow f = X \hat{w}_{MAP} = \begin{bmatrix} -2.013 \\ -1.007 \\ +1.007 \\ +2.013 \end{bmatrix}$

- curvature of the model

compute the diagonal variance vector

$V_n = \sigma(\hat{f}_n)[1 - \sigma(\hat{f}_n)]$:

h	1	2	3	4
$\hat{f}_n$	-2.013	-1.007	1.007	2.013
$\sigma(\hat{f}_n)$	0.117	0.267	0.733	0.883
v_n	0.103	0.196	0.196	0.103

plugging into hessian

$$H = -X^T \operatorname{diag}(v) X - I = \begin{bmatrix} 1.59 & 0 \\ 0 & 2.22 \end{bmatrix}$$

- Laplace Gaussian

$$q(w) = \mathcal{N}(\hat{w}_{MAP}, S) \qquad S = (-H)^{-1} = \begin{bmatrix} 0.625 & 0 \\ 0 & 0.449 \end{bmatrix}$$

Approximating the posterior predictive distribution

- choosing a test input, $x^* = \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$ (intercept + a raw $x^* = 1.5$)

- compute
  - $m^* = m^T x^* = [0 \quad 1.00] \begin{bmatrix} 1 \\ 1.5 \end{bmatrix} = 0 \cdot 1 + 1 \cdot 1.5 = \underline{\underline{1.5}}$ (map)
  - $v^* = x^{*T} S x = [1 \quad 1.5] \begin{bmatrix} 0.6250 & 0 \\ 0 & 0.4497 \end{bmatrix} \begin{bmatrix} 1 \\ 1.5 \end{bmatrix} = 0.6250 + 1.5^2 \cdot 0.4497 = 1.6369$

  these two scalars summarise everything we need from the entire Gaussian posterior

Plug-in (Map) approximation

$$\boxed{P(y^* = 1 \mid y, x^*) \approx \sigma(m^*) = \frac{1}{1 + e^{-1.5099}} = 0.819}$$

↳ we freeze the uncertainity in w, pretend $w = \hat{w}_{MAP}$ is true, then use the clasical logistic link.

Probit Approximation

A logistic curve $\sigma(z)$ can be matched by probit curve $\Phi(\lambda z)$ if $\lambda \approx \sqrt{\pi/8}$. The deterministic replacement let's us to do Gaussian integral in closed form because $\Phi(\cdot)$ is the CDF of a normal and the product of the two normals is still a scaled normal.

$$\lambda = \sqrt{\pi/8} = 0.6267 \; ; \quad \frac{m^*}{\sqrt{\frac{8}{\pi} + v^*}} = \frac{1.5099}{\sqrt{1.273 + 1.6369}} = \frac{1.5099}{1.6388} = 0.9215 \Rightarrow P_{probit} = \Phi(0.9215) = 0.770$$

- interpretation: the denominator inflates the effective standard deviation by $\sqrt{8/\pi}$ before adding v^*; this correction shrinks the probability towards 0.5 to reflect weight uncertainity

Monte Carlo estimator

- Generate $S = 10{,}000$ draws of f^*, because f^* is linear in w

$$f^* \sim \mathcal{N}(m^*, v^*) = \mathcal{N}(1.5099, 1.6369)$$

- draw 10.000 i.i.d. samples $f^*_{(i)}$ from that normal
- Push each sample through the sigmoid

Compute $\sigma(f^*_{(i)})$ and average:

$$P_{MC} = \frac{1}{10\,000} \sum_{i=1}^{10000} \sigma(f^*_{(i)}) \approx 0.763$$

- standard error $\sqrt{\frac{P(1-P)}{S}} \approx 0.004$, so ± 2 band is roughly $[0.755, 0.771]$

Comparison:

- Plug-in is the most confident as it ignores the uncertainty (v^*)
- Probit ∧ MC are very close. MC = Golden standard, but probit is cheap.

Summary

- m^* is the best Guess logit for x^*. if $m^* > 0$, class 1 is more likely on average
- V^* encodes how unsure we are about that logit

Three Approximation Methods

- Plug-in: ignore weight uncertainty: just use one weight vector (the MAP)
- monte Carlo: Sample many weights from posterior, average sigmoid $(x \cdot w)$
- Probit appror: Replace logistic link w/ a probit and do the Gaussian integral in closed form.

All 3 start from $f_* = x_*^T w$ $\quad w \sim \mathcal{N}(m, S)$, the diff. are purely in how they treat the link func.
$p(y_* = 1 | f_*) = \sigma(f_*)$

Task 6.1 Show that the pos. pred. prob $p(y^* = 1 | y, x) \to 0.5$ as variance $v^* \to \infty$

- in probit you compute

$$p = \Phi(z), \quad z = \frac{m_*}{\sqrt{8/\pi + v^*}}$$

$$\lim_{v^* \to \infty} \frac{m^*}{\sqrt{\frac{8}{\pi} + v^*}} = 0 \Rightarrow \Phi(0) = 0.5$$

where $m_* = x_*^T m$ ∧ $v_* = x_*^T S x_*$

For large v_*:

$$z = \frac{m_*}{\sqrt{v_*}} \sqrt{\frac{1}{1 + 8/(\pi v_*)}} \to 0$$

The standard norm. CDF satisfies $\Phi(0) = 0.5$ ∴ $p = \Phi(z) \to 0.5$

When post. var. of the margin f_* is HUGE, we are unsure whether the latent var. is pos. or neg. so the best calibrated prob. assignment is a coin flip (0.5)